

Week 11 — Turtle Geometry

Name: ____ Date: ____

The turtle turns the **outside** of every corner — and one full lap around any polygon is always **360°** of turning.
Turn at each corner = $360 / n$.

Today's words

Word	What it means
turtle	Python's drawing robot — it draws everywhere it walks
forward(n) / right(d)	walk n steps / spin d degrees (turning never draws)
penup() / pendown()	lift the pen to move invisibly / put it back down
exterior angle	the turn at a corner — the outside; what the turtle turns
interior angle	the angle inside the corner — always 180 – the turn

1 • Be the turtle 🐢

Your pencil is the pen. Start at **S**, facing →. One grid square = 20 steps. Draw the turtle's path.

```
turtle.forward(60)
turtle.right(90)
turtle.forward(40)
turtle.right(90)
turtle.forward(60)
```

```
S . . . . .
. . . . .
. . . . .
. . . . .
```

What familiar shape is it — and which side is missing? ____

2 • The polygon table 📐

Fill in the missing values. (Turn = $360 / n$. Interior = $180 - \text{turn}$.)

sides n	turtle's turn (exterior)	interior angle
3	_____	60°
4	90°	90°
5	72°	_____
6	_____	_____
8	_____	_____

3 • Spot the bug 🐛

This was supposed to draw a square. Circle the bug, then answer below.

```
import turtle
for side in range(4):
    turtle.forward(100)
    turtle.right(90)
```

What does it draw instead? ____

🧠 Brain teaser (optional — take it home)

Write the turtle commands to draw a **rectangle 200 steps wide and 100 steps tall** — or the **first letter of your first name**. Rules: only `forward()`, `backward()`, `right()`, `left()`, `penup()`, and `pendown()`, one command per line. Trace it with your finger to test — you are the turtle. Bring it next week for a shout-out (we may run it on the smartboard!).